



Lab 02: Set Up Application Load Balancers

Google Cloud Associate Cloud Engineer | GSP155

Overview

Set up a Layer 7 (L7) Application Load Balancer on Google Cloud using Compute Engine VMs. Unlike the Network Load Balancer (Lab 01), an L7 ALB understands HTTP/HTTPS protocols and can make routing decisions based on URL, headers, cookies, and request content.

Key Difference from Lab 01 (NLB vs ALB)

Network Load Balancer (Lab 01)	Application Load Balancer (Lab 02)
Layer 4 — IP/Port based	Layer 7 — HTTP/HTTPS content aware
Target pools	Managed instance groups
Cannot route by URL	Can route by URL, headers, cookies

Objectives

- Configure the default region and zone
- Create 3 web server VM instances with Apache
- Create an instance template and managed instance group
- Configure an Application Load Balancer
- Test traffic distribution to instances

Task 1 — Set Default Region & Zone

```
gcloud config set compute/region [YOUR-REGION]
gcloud config set compute/zone [YOUR-ZONE]
```

```
student_03_6446d07d57a2@cloudshell:~ (qwiklabs-gcp-03-78618591972b)$ gcloud config
set compute/region us-east1
Updated property [compute/region].
student_03_6446d07d57a2@cloudshell:~ (qwiklabs-gcp-03-78618591972b)$ gcloud config
set compute/zone us-east1-c
Updated property [compute/zone].
student_03_6446d07d57a2@cloudshell:~ (qwiklabs-gcp-03-78618591972b)$
```

Task 2 — Create Web Server Instances

Created 3 VMs (www1, www2, www3) with Apache installed and a unique homepage each.
Firewall rule added to allow HTTP traffic on port 80.

```
gcloud compute instances create www1 --zone=[ZONE] --tags=network-lb-tag --machine-type=e2-small --image-family=debian-11 --image-project=debian-cloud
gcloud compute instances create www2 --zone=[ZONE] --tags=network-lb-tag --machine-type=e2-small --image-family=debian-11 --image-project=debian-cloud
gcloud compute instances create www3 --zone=[ZONE] --tags=network-lb-tag --machine-type=e2-small --image-family=debian-11 --image-project=debian-cloud
gcloud compute firewall-rules create www-firewall-network-lb --target-tags network-lb-tag --allow tcp:80
```

```
DISABLED: False
student_03_6446d07d57a2@cloudshell:~ (qwiklabs-gcp-03-78618591972b) $ gcloud compute instances list
NAME: www1
ZONE: us-east1-c
MACHINE_TYPE: e2-small
PREEMPTIBLE:
INTERNAL_IP: 10.142.0.2
EXTERNAL_IP: 35.229.20.223
STATUS: RUNNING

NAME: www2
ZONE: us-east1-c
MACHINE_TYPE: e2-small
PREEMPTIBLE:
INTERNAL_IP: 10.142.0.3
EXTERNAL_IP: 34.24.120.110
STATUS: RUNNING

NAME: www3
ZONE: us-east1-c
MACHINE_TYPE: e2-small
PREEMPTIBLE:
INTERNAL_IP: 10.142.0.4
EXTERNAL_IP: 34.139.28.162
STATUS: RUNNING
student_03_6446d07d57a2@cloudshell:~ (qwiklabs-gcp-03-78618591972b) $
```

```
student_03_6446d07d57a2@cloudshell:~ (qwiklabs-gcp-03-78618591972b) $ curl http://35.229.20.223
Web Server: www1
student_03_6446d07d57a2@cloudshell:~ (qwiklabs-gcp-03-78618591972b) $ curl http://34.24.120.110
Web Server: www2
student_03_6446d07d57a2@cloudshell:~ (qwiklabs-gcp-03-78618591972b) $ curl http://34.139.28.162
Web Server: www3
student_03_6446d07d57a2@cloudshell:~ (qwiklabs-gcp-03-78618591972b) $
```

Task 3 — Create the Application Load Balancer

The ALB requires more setup than the NLB. It uses an instance template, a managed instance group, a backend service, a URL map, an HTTP proxy, and a forwarding rule.

Step 1 — Create instance template:

```
gcloud compute instance-templates create lb-backend-template --region=[REGION] --  
network=default --subnet=default --tags=allow-health-check --machine-type=e2-medium  
--image-family=debian-11 --image-project=debian-cloud
```

Step 2 — Create managed instance group:

```
gcloud compute instance-groups managed create lb-backend-group --template=lb-  
backend-template --size=2 --zone=[ZONE]
```

Step 3 — Create firewall rule for health checks:

```
gcloud compute firewall-rules create fw-allow-health-check --network=default --  
action=allow --direction=ingress --source-ranges=130.211.0.0/22,35.191.0.0/16 --  
target-tags=allow-health-check --rules=tcp:80
```

Step 4 — Reserve global static external IP:

```
gcloud compute addresses create lb-ipv4-1 --ip-version=IPv4 --global
```

Step 5 — Create health check:

```
gcloud compute health-checks create http http-basic-check --port 80
```

Step 6 — Create backend service:

```
gcloud compute backend-services create web-backend-service --protocol=HTTP --port-  
name=http --health-checks=http-basic-check --global
```

Step 7 — Add instance group to backend:

```
gcloud compute backend-services add-backend web-backend-service --instance-  
group=lb-backend-group --instance-group-zone=[ZONE] --global
```

Step 8 — Create URL map:

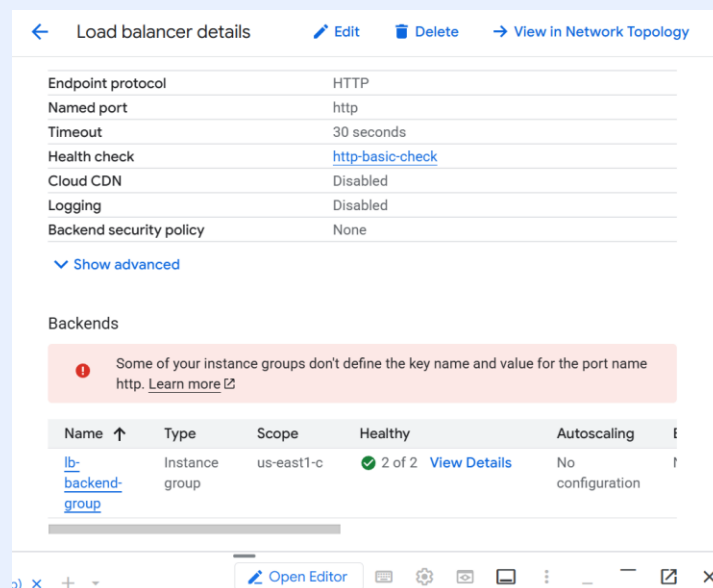
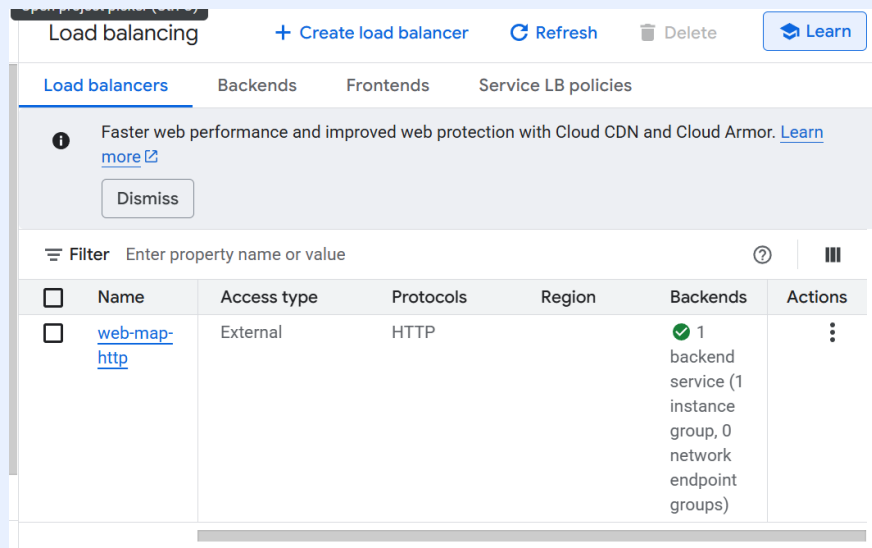
```
gcloud compute url-maps create web-map-http --default-service web-backend-service
```

Step 9 — Create HTTP proxy:

```
gcloud compute target-http-proxies create http-lb-proxy --url-map web-map-http
```

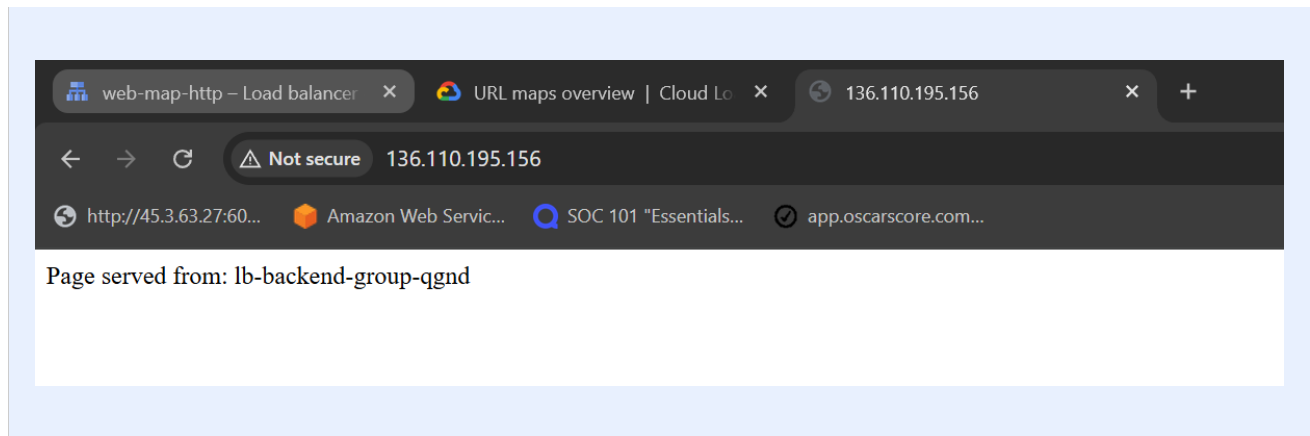
Step 10 — Create forwarding rule:

```
gcloud compute forwarding-rules create http-content-rule --address=lb-ipv4-1 --  
global --target-http-proxy=http-lb-proxy --ports=80
```



Task 4 — Test Traffic to Instances

Opened the load balancer IP in a browser. The page showed which backend instance served the request (e.g. Page served from: lb-backend-group-xxxx) confirming traffic distribution is working.



Real World Application

An Application Load Balancer is used by companies like Google, Netflix, and Airbnb to route traffic intelligently. For example:

- `yoursite.com/videos` → routes to video streaming servers
- `yoursite.com/api` → routes to API backend servers
- `yoursite.com/images` → routes to Cloud Storage

This is impossible with a Network Load Balancer (Lab 01) because it cannot read URL content — it only sees IP addresses and ports.

What I Learned

- A Layer 7 ALB understands HTTP content and can route by URL, headers, and cookies
- Managed instance groups allow autoscaling and autohealing of backend VMs
- Instance templates define the configuration for all VMs in a group
- Health checks ensure only healthy instances receive traffic
- URL maps allow routing different paths to different backend services
- The ALB setup has more components than NLB but offers much more flexibility

Resources

GCP Application Load Balancer Docs: <https://cloud.google.com/load-balancing/docs/application-load-balancer>
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